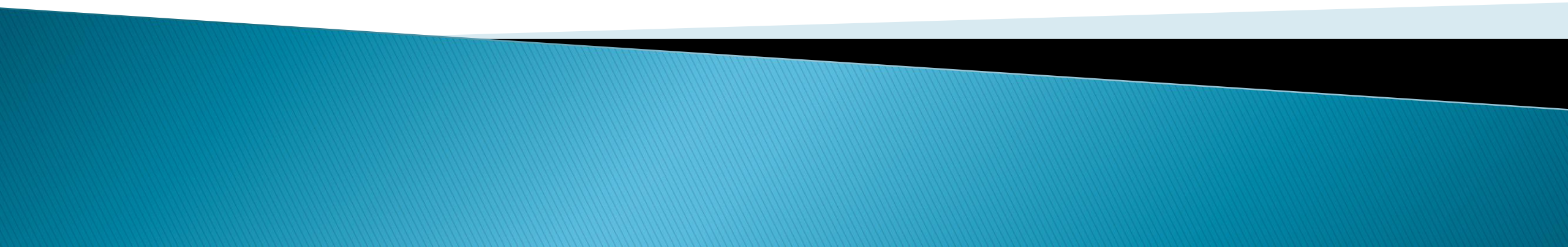


How RDD solves the problem ?

by Priti Bhardwaj , CDAC NOIDA, India

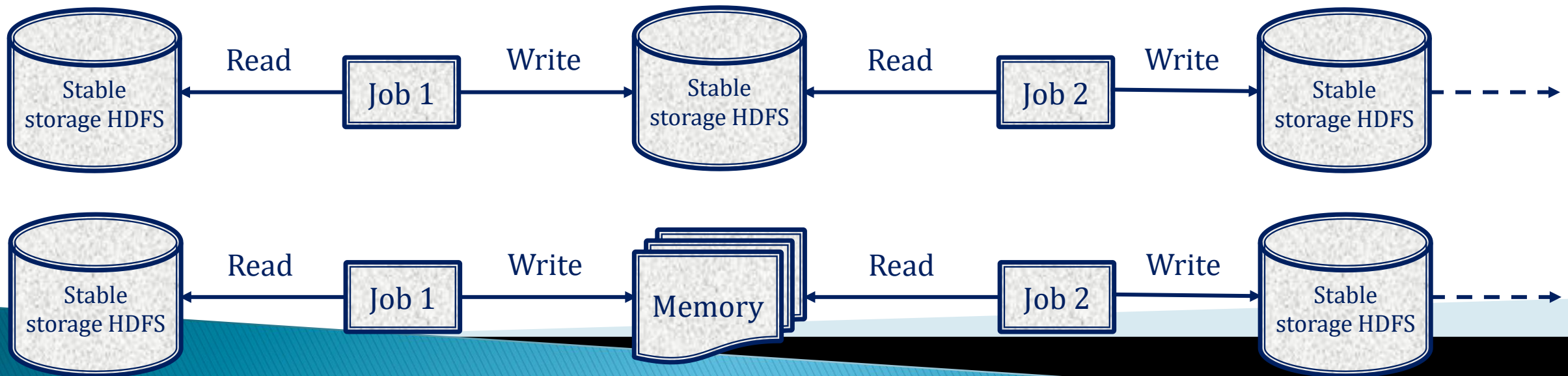


How Spark RDD solve the problem?

- There Exists a Synchronization Barrier between Map and Reduce Tasks
- Data Sharing is slow with Hadoop MapReduce - Spark RDDs solve this.

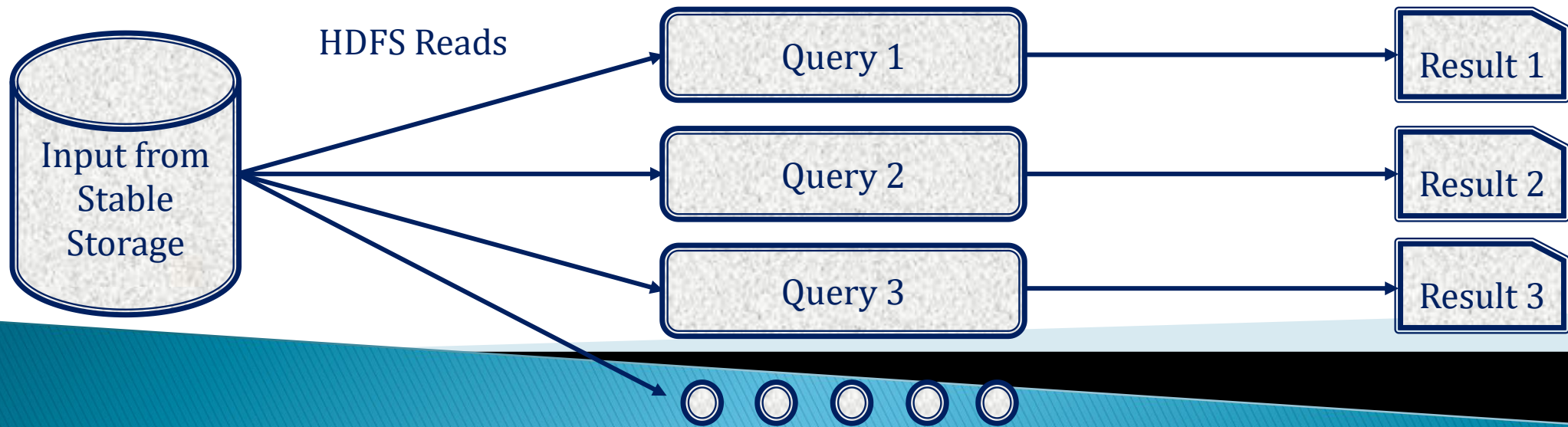
How Spark RDD solve the problem?

- Iterative Operations on Hadoop MapReduce and on Spark RDDs



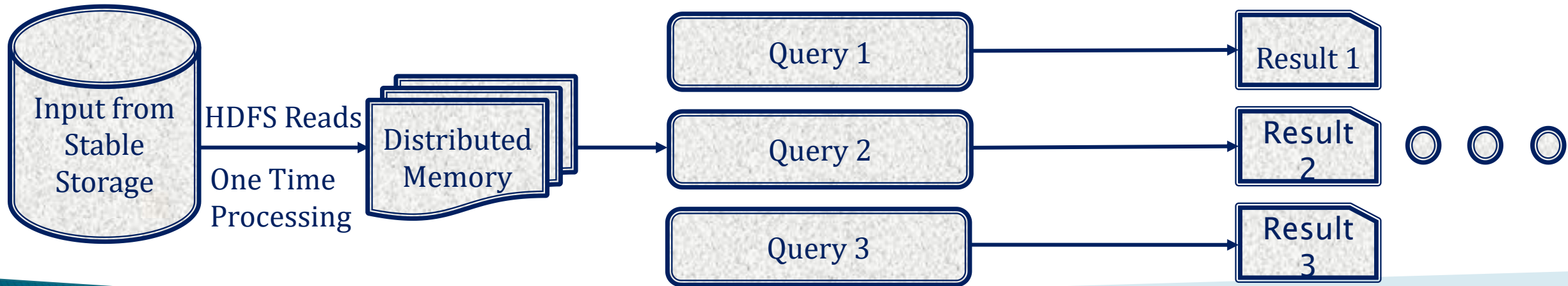
How Spark RDD solve the problem?

➤ Interactive Operations on Hadoop MapReduce



How Spark RDD solve the problem?

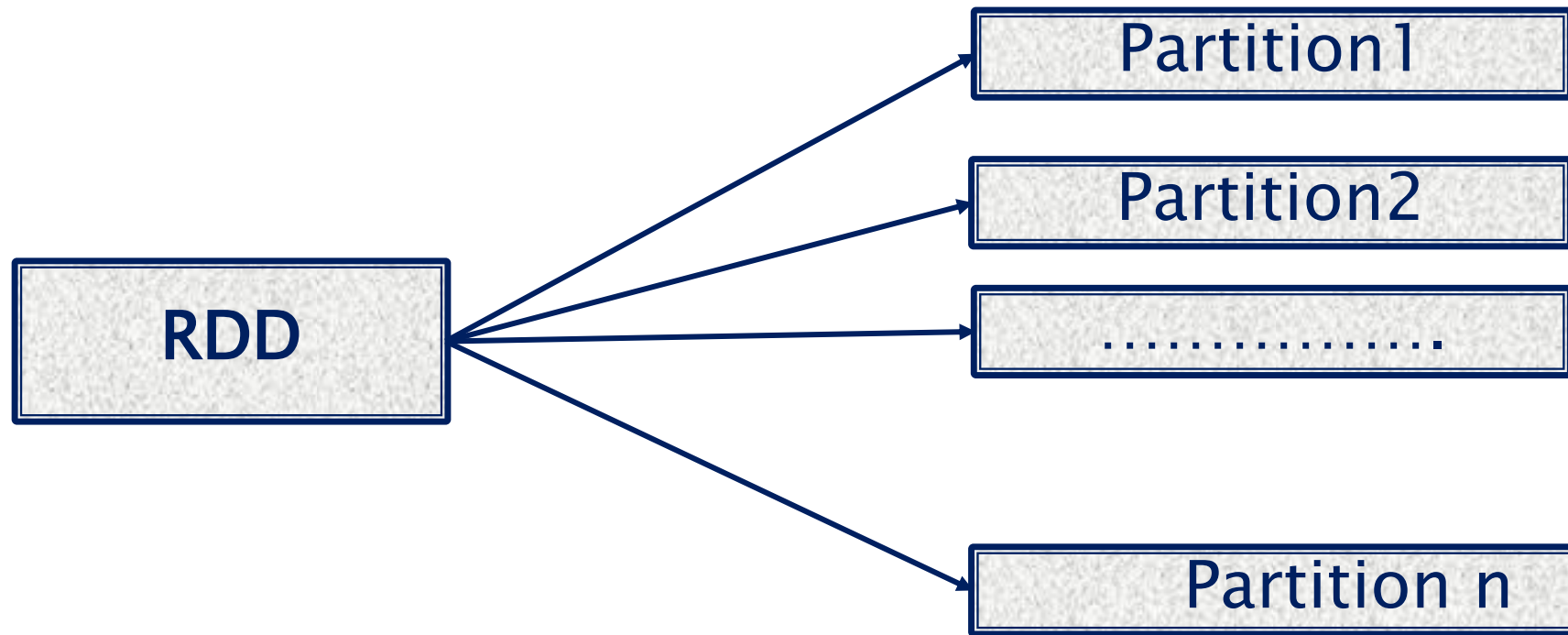
➤ Interactive Operations on Spark RDDs



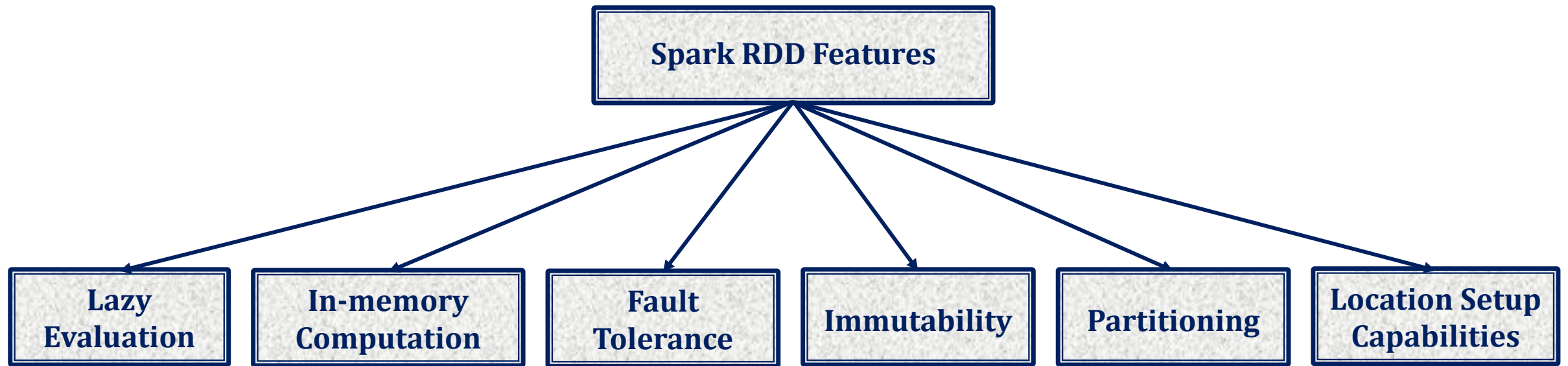
What is Spark RDDs?

- RDD are the building blocks of any spark application.
- **Resilient:** Fault tolerant and capable of rebuilding data on failure.
- **Distributed:** Distributed data among the multiple nodes in a cluster.
- **Dataset:** Collection of partitioned data with values.
- Each dataset in RDD is divided into logical partitions.

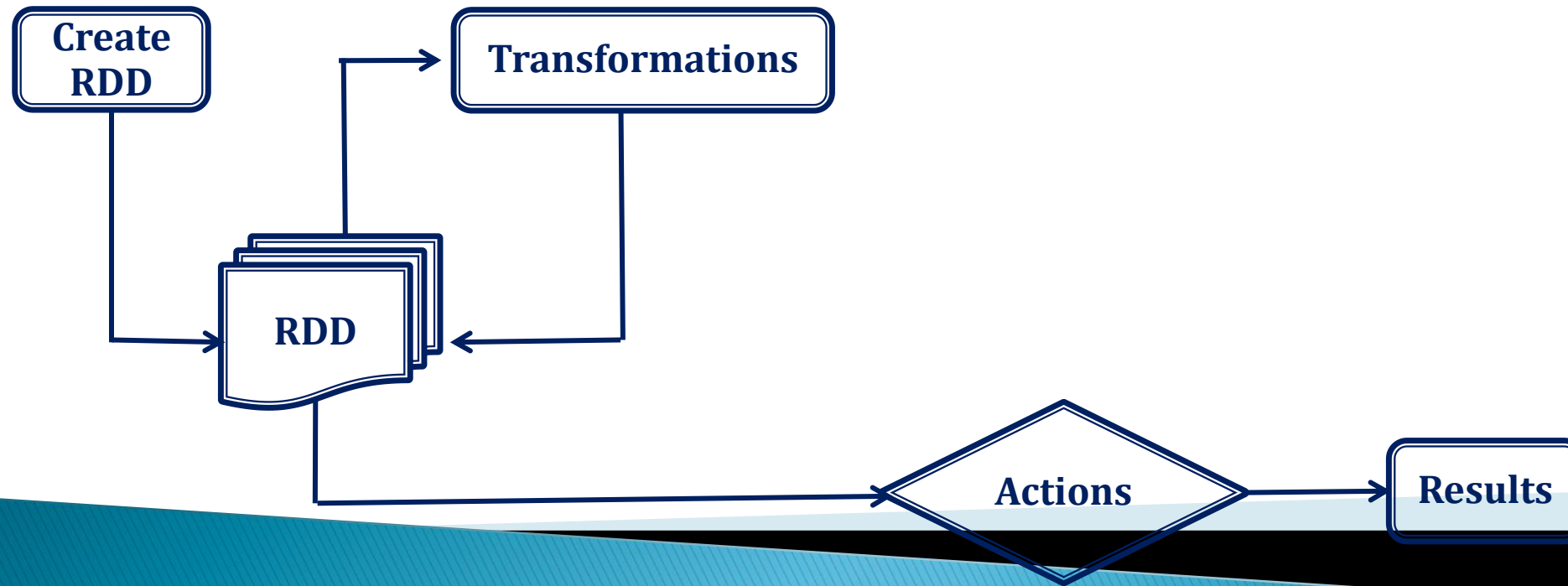
Spark RDD



Spark RDD Features



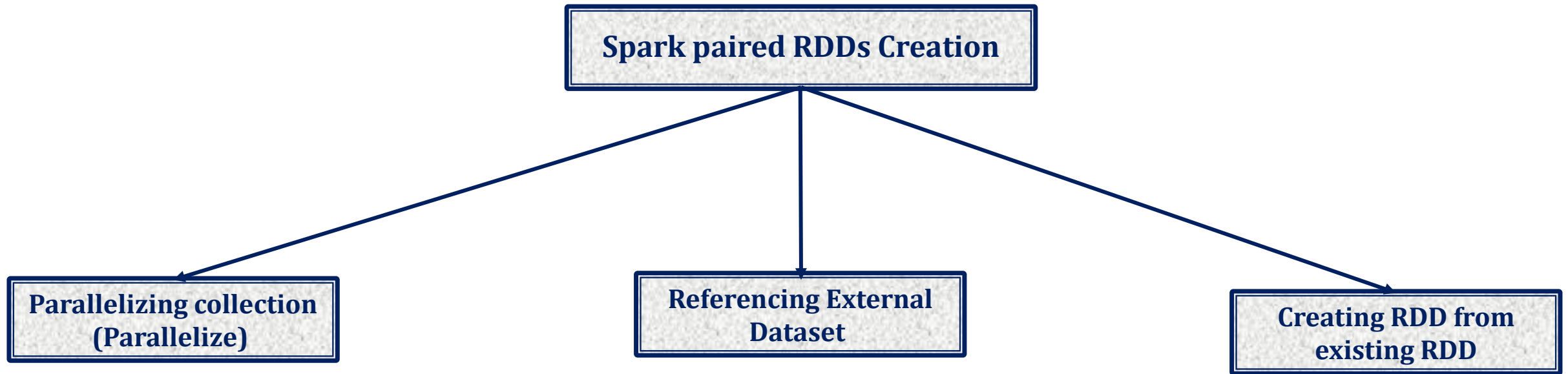
Spark RDD Operations



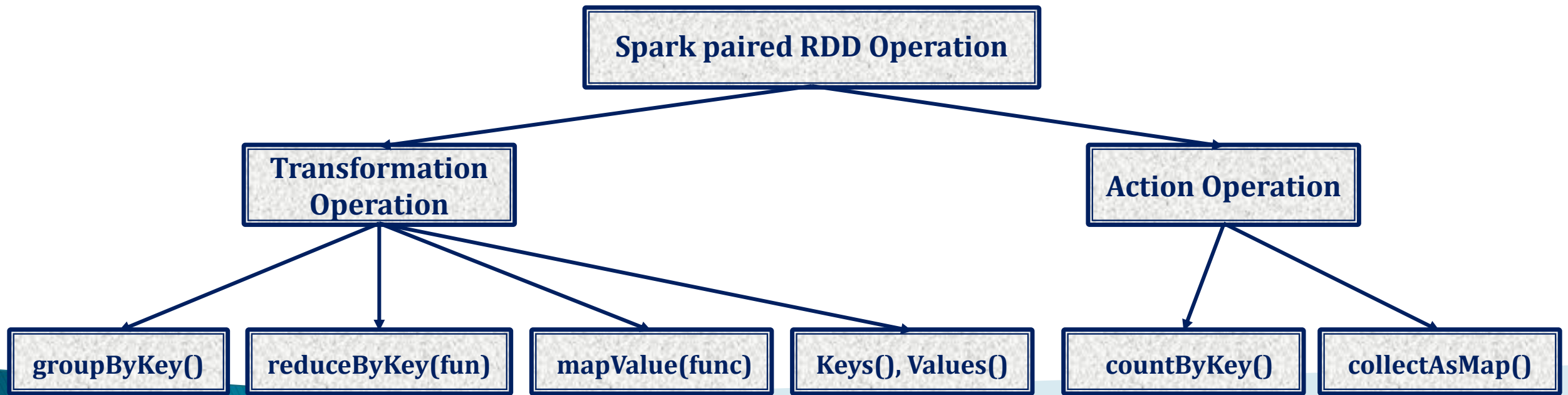
Introduction to Spark paired RDDs

- Spark Paired RDDs contains a key-value pair.
- Special operations like shuffle, grouping or aggregating elements by key.
- Operations for the key-value pair are available in the Pair RDD functions class.
- We can regroup the data across the network.
- Operations like `reduceByKey()` ,`join`.

How to create Spark paired RDDs



Spark paired RDD Operation



THANK YOU